5.3.2 Operating conditions at power-up/power-down

The parameters given in the table below are derived from tests performed under the ambient temperature condition summarized in Table 17.

Table 18. Operating conditions at power-up/power-down

Symbol	Parameter	Min	Max	Unit
t_{VDD}	V_{DD} rise-time rate	0	∞	µs/V
	V_{DD} fall-time rate	0	∞	ms/V

5.3.3 Embedded reset and power control block characteristics

The parameters given in the table below are derived from tests performed under the ambient temperature conditions summarized in Table 17.

Table 19. Embedded reset and power control block characteristics

The values in this table are evaluated by characterization - Not tested in production, unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{RSTTEMPO}$(1)(2)	Reset temporization after POR released	V_{DD} rising	-	-	463	µs
$V_{POR/PDR}$	Power-on/power-down reset threshold (BORH_EN =0)	Rising edge	2.56	2.61	2.64	V
		Falling edge	2.53	2.58	2.61	
V_{PVD}	Programmable Voltage Detector threshold	Rising edge	3.00	3.04	3.08	
		Falling edge	2.89	2.93	2.96	
$V_{hyst\_POR\_PDR}$	Hysteresis for power-on/power-down reset	-	-	30	-	mV
V_{hyst_PVD}	Hysteresis voltage of PVD	-	-	110	-	
I_{DD_PVD}(2)	PVD consumption from VDD	-	-	-	0.63	µA

1. *Specified by design - Not tested in production.*
2. *From POR threshold crossing to NRST pull-up resistor activation.*

5.3.4 Inrush current and inrush electric charge characteristics

The parameters provided in the following table are specified by design simulation and are not tested in production.

Table 20. Embedded internal voltage reference

The typical values are provided for V_{DD} = 3.3V and for a typical decoupling capacitor value .
The product consumption on V_{DDCORE} is not included in the inrush current and inrush electric charge

Symbol	Parameter	Typ	Unit
I_{RUSH}	Inrush current during voltage regulator power-on (POR) or wake-up from standby	35	mA
Q_{RUSH}	Inrush electric charge during voltage regulator power-on (POR) or wake-up from standby.	2.8	µC